

Rowan SAME

Search And Rescue Drone



Prepared For: SAME Philadelphia

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1. EXECUTIVE SUMMARY

The Rowan SAME Search and Rescue drone project is an engineering project with the goal to build a high performance drone at an affordable price point. In the existing market, municipalities and emergency responders are required to spend \$10,000 or more on a professional drone or use consumer grade drones that have a limited payload capacity or flight duration.

Our mission is to close this gap by designing and manufacturing a drone that is able to maintain a 2 hour flight time. By using a lightweight carbon-fiber frame and a high capacity power system, we plan to achieve long flight durations, be able to carry a 3.5 kg payload for vital missions, and remain affordable for real world rescue work.

Over the next 15 weeks, the team will go through the engineering cycle. The team plans to produce CAD models for the components as well as a full assembly. We also plan to conduct stress testing and FEA on the drone. In the final stage, field testing will occur. By the end of this project, we aim to have a professional search and rescue drone that can be made accessible to municipalities at a fraction of current market costs.

2. PROBLEM STATEMENT & OBJECTIVES

2.1 Current State

The current Search and Rescue drone market currently has three main limitations that prevent widespread adoption and use by local emergency teams:

- **Financial Barrier:** Professional Search and Rescue drones often cost upwards of \$10,000. This price makes these drones inaccessible to smaller municipalities looking to purchase them.
- **Payload Capacity:** Most drones are designed to be able to support their cameras, not extra cargo. They lack the motor torque required to carry the most basic medical supplies without the possibility of failing.
- **Range:** The average battery life of search and rescue drones is roughly 30-50 minutes. This is not enough to search a large scale forest or disaster area.

2.2 Target State

The target state for the Rowan SAME drone project is to create a low cost search and rescue drone that can be easily deployed by rescue workers and deliver life saving medical supplies all on one charge.

2.3 Key Technical Objectives

- **Objective 1 (Load):** Static load capacity of 3.5 kg with a FoS of 1.5.
- **Objective 2 (Endurance):** Must be able to sustain 2-hour long missions at a time.
- **Objective 3 (Altitude):** Shall be able to reach and sustain an altitude of no more than 400 ft. AGL as outlined in FAA Part 107.
- **Objective 4 (Range):** Must be equipped with long range receiving and transmitting capabilities.
- **Objective 5 (High Definition Optical System):** Capable of tracking small objects from the desired altitude.
- **Objective 6 (Modularity):** Interchangeable sensors suitable for different types of missions and ease of transportation.

3. PROPOSED TECHNICAL SOLUTION

The proposed solution utilizes analytical and simulation approaches to minimize development costs, and create a successful prototype first attempt. Using well established aircraft design rules and modern technologies like CAD and fluid simulation softwares, it is easier than ever before to produce a low cost prototype without major design flaws, and smoothly move to a final design.

3.1 Assembly Architecture

- **Fuselage:** Woven Carbon Fiber Unibody, casted as one piece for weight reduction and increased strength and stiffness. Houses avionics, search equipment, power source and motor. It will be outfitted with custom modular sensor packages for different types of missions. Modular design allows for more compact size and ease of transportation.
- **Empennage Assembly:** A Carbon Fiber rod replacing a standard, bulky and complex empennage for a strong, lightweight and cost effective solution. Attached V-tail with symmetrical airfoil **ruddervators** combine elevator and rudder into one piece which requires less actuators and increases the simplicity of the design.
- **Wing:** Split into two components, its standard and widely known for its great aerodynamic properties NACA 2412 provides a lot of lift at low speeds, suitable for slow, long and around a point search missions.

3.2 Fabrication Methodology

- **Prototyping:**

FDM, SLA and laser cutting manufacturing techniques will be widely used in the process of initial low cost prototyping. These methods are quick and cheap, and every design change can be printed and tested before a final, more expensive part is produced. This not only saves money, but also improves efficiency, as modern 3D printers are extremely fast and reliable, allowing design teams to confirm their design within hours.

- **Production:**

First Production/ Final Prototype model will be fully made of composite, such as a woven carbon fiber fabric. It is an optimal choice for strength-to-weight ratio, an important metric in aerospace products. A drawback related to this choice of material is a difficulty in manufacturing, as well as specialized tools which are needed in this process. Fully composite shell allows this platform to operate in virtually any atmospheric conditions, as carbon fiber is highly resistant towards adverse weather. This is important, because search operations need to be performed in all kinds of weather, or locations like by the ocean shore where ocean breeze is rich in salt and could cause corrosion on a standard fuselage in the long term.

4. TECHNICAL SPECIFICATIONS

Feature	Requirement	Tolerance/Metric
Material Grade	Woven Carbon Fiber (2x2 Twill)	N/A
Operating Pressure	101.3 kPa (Sea Level)	±5%
Surface Finish	Gloss, UV-Resistant Clear Coat	Ra < 0.8 µm
Maximum Safety Factor	1.5	Yield Strength Basis
Thermal Range	-20°C to 60°C	Component Standard
Max Flight Duration	120 Minutes (2 Hours)	At 20 m/s Cruise
Propulsion System	Pusher Propeller	T-Motor AT4130 / 15" Prop
Navigation Sensors	Airspeed (Pitot), Lidar, GNSS	Triple Redundant GNSS
Camera System	Dual-Axis FPV + Thermal (FLIR/Sony)	1080p 60fps / 640px Thermal
Communication Range	50 – 60 km	LoS (Line of Sight)
Primary Flight Computer	Cube Orange+ (ArduPilot/PX4)	Triple Redundant IMU
Secondary Computer	Nvidia Jetson Orin Nano	Edge AI / CV
Wingspan	7.0 ft (2.13 m)	High-Aspect Ratio
Maximum Altitude	400 ft AGL	Hard-coded Geo-fence
Max Take-Off Weight	9.5 kg	Including 3.5 kg Payload

5. TESTING & VALIDATION

To ensure the drone is safe and reliable for search and rescue missions, the team will conduct testing throughout the engineering cycle of the drone. We will use both digital and physical testing methods.

5.1 Simulations & FEA

- **Simulations:** The team will use industry standard tools such as Simulink, CAD, and other softwares to simulate the drone. We will be able to ensure that the drone is able to stay stable during payload release or severe weather conditions.
- **Finite Element Analysis (FEA):** The team will perform stress and strain analysis on the CAD model of the airframe. This will allow us to identify weak points in the frame and ensure the structure can maintain a Factor of Safety (FOS) of 1.5.

5.2 Physical Testing

- **Power System Thermal Test:** Since the drone will have a 2-hour flight time, the team will test the batteries to make sure that the internal components won't overheat during flight.
- **Center of Gravity Test:** With the medical supplies loaded in the drone, the team will determine the best position for them to make sure the drone is stable and efficient.
- **Thrust & Efficiency Test:** The team will determine a proper speed for the motor that covers the most amount of area while using the least amount of battery to achieve our desired 2-hour flight time.

5.3 Flight Trials

- **Stability Test:** Initial flights will be done at low altitudes. We will need to verify that the airfoil and motor provides enough lift for the drone before flying at high altitudes or long distances.
- **Payload Test:** The team will test the release of a dummy medical supplies payload. This will show us if the sudden weight change within the drone causes any stability issues as well as ensure the supplies aren't damaged on impact.
- **Endurance Test:** The team will perform a flight test to ensure the drone can stay in the air for the 2-hour target duration while carrying the full 3.5 kg payload.

6. Bill of Materials

Airframe and Mold Fabrication	Cost
Uni-directional and 2x2 Twill for wing skins and spars	\$300
Rohacell or Divinycell foam for sandwich construction	\$150
CNC-machined tooling board or high-density epoxy tooling resin	\$200
Peel ply, breather, and sealant	\$100
Propulsion System	
T-Motor AT-Series (e.g., AT4130) or equivalent high-efficiency outrunner	\$180
Castle Creations Talon 90 (or similar with a high-amperage BEC)	\$120
3x Sets of high-efficiency folding or fixed carbon props	\$90
Custom CNC aluminum or heavy-duty composite mount	\$60
Flight Control and Custom Avionics	
Cube Orange+ with ADS-B Carrier Board	\$450
Here4 Multiband GNSS	\$230
Custom PCB Components: Logic level shifters, power filters, and ESP32/Teensy	\$40
Actuators and Linkages	
4x MKS or Blue Bird high-torque, metal gear digital servos	\$320
Carbon fiber pushrods, titanium turnbuckles, and ball links	\$60
Reconnaissance Payload and Sensors	
RunCam Phoenix 2 (for low-latency FPV)	\$50
Matek Digital Pitot Tube	\$60
Sony RX0 II	\$350
Nvidia Jetson (for image processing and data logging)	\$120
Long Range Communication	
DragonLink V3 Slim	\$350
Walksnail Avatar HD Pro	\$270
Power Source	
(72) Molicel P42A 18650 Cells	\$200
6S 100A Smart BMS	\$60
0.2mm Pure Nickel Strips and Kapton Tape	\$30
Capital Tooling	

Sovol SV03/Similar Large Format 3D Printer for mold plugs	\$300
Kweld or equivalent Battery Spot Welder	\$150
Consumables	
3kg ASA and CF-Nylon Filament	\$120
West Systems Epoxy/Hardener	\$80
Sanding/Shop supplies	\$100
RESERVE/MISC	
Buffer for Shipping, Duty, and Fastener Hardware.	\$100
TOTAL: \$4,640	

7. PROJECT MILESTONES

Phase	Task Description	Estimated Duration
I	CAD Modeling & FEA Stress Analysis	Week 1–5
II	Material Procurement & Tooling Setup	Week 5–7
III	Prototype Machining & Assembly	Week 7–10
IV	Destructive/Non-Destructive Testing (NDT)	Week 10–12
V	Final Finishing & Installation	Week 12–15

8. RISK MITIGATION & REGULATORY COMPLIANCE

Risk Category	Potential Impact	Mitigation Strategy
Structural Fatigue	Carbon fiber delamination at wing root or high-stress corners.	Apply 5mm fillets to redistribute stress; perform NDT (Non-Destructive Testing) after every 5 flight hours
Thermal Runaway	High-capacity LiPo batteries overheating during 2-hour missions.	Integrated heat sinks in the fuselage airflow path; real-time telemetry alerts for battery temps > 55°C.
RF Interference	Loss of C2 (Command & Control) link in urban search areas.	Implementation of DragonLink V3 (433MHz) for superior penetration and automatic "Return to Launch" (RTL) on signal loss.
Payload Shift	Center of Gravity (CoG) instability during medical supply release.	Rail-mounted internal payload system to ensure the weight drops vertically from the neutral CoG point.